Bequests, Fertility and Barriers to Entrepreneurship:

The Intergenerational Transmission of Economic Inequality

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Abstract

This paper studies how parents' fertility and bequest decisions jointly shape their children's access to entrepreneurship under borrowing constraints. While the existing literature on intergenerational inequality emphasizes bequests, it largely abstracts from how fertility choices dilute per-child bequests and thereby affect entrepreneurial opportunities. I develop an overlapping-generations model with endogenous fertility, bequests, and occupational choices, calibrated to U.S. data. The model shows that fertility-induced dilution of per-child bequests interacts with borrowing constraints to distort entrepreneurial entry and generate aggregate misallocation. Eliminating borrowing constraints in the U.S. raises income per capita by 13 percent but disproportionately benefits high-productivity firms, whose expansion raises the equilibrium interest rate and crowds out marginal entrepreneurs, while higher asset returns increase inequality and reduce intergenerational mobility.

Keywords: Entrepreneurship, Financial Barriers, Bequest, Fertility

JEL Classification: O16, E24

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1 Introduction

Entrepreneurship is crucial for economic development (Schumpeter, 1934), but many entrepreneurs face borrowing constraints that limit their ability to start or grow businesses (Buera, 2008; Evans and Jovanovic, 1989). Intergenerational transfers can help alleviate financial constraints, and existing work has therefore emphasized their importance as determinants of entrepreneurial opportunity (Holtz-Eakin et al., 1994). However, current literature has largely abstracted from parental fertility choices, which determine resources per child. Empirical evidence has shown that wealthier households tend to have fewer children (Becker, 1960; Borg, 1989; Docquier, 2004; Jones et al., 2008) and can concentrate more resources and leave larger bequests to each child. Individuals with high-income parents typically encounter fewer financial barriers and are more likely to become entrepreneurs. Ignoring the endogenous fertility channel would bias the estimated influence of financial barriers on economic growth and inequality.

This paper examines how parents' decisions regarding fertility and bequests impact their children's ability to become entrepreneurs when financial markets are imperfect. Fertility determines how parental resources are diluted across children, and financial barriers make inherited resources matter for entrepreneurial entry and firm scale. Thus, parental bequests and fertility choices interact with borrowing constraints to distort children's entrepreneurship decisions. Highly talented individuals from larger, less wealthy families may be unable to obtain enough capital to operate optimally, whereas less talented individuals from smaller, wealthier families may be able to enter and expand. This mechanism generates a mismatch between entrepreneurial talent and capital, reducing aggregate efficiency. Because entrepreneurship is a key channel for wealth accumulation and many of the wealthiest individuals are business owners (Cagetti and De Nardi, 2006), these distortions can also amplify income inequality and reduce intergenerational mobility.

To study these dynamics, I develop an overlapping generations model where parents choose how many children to have and how much wealth to transfer to their children. There is no heterogeneity in bequests within the family, and all children receive the same average bequest. Children inherit parental bequests as assets and partially inherit entrepreneurial talent following an AR(1) process. In a benchmark economy without financial barriers, individuals can become entrepreneurs as long as their entrepreneurial talent is high enough, and all entrepreneurs operate at their optimal scale. I then introduce financial barriers for entrepreneurship arising from limited contract enforceability, following Buera et al. (2011), which generates endogenous borrowing constraints that depend on individuals' wealth and entrepreneurial talent. In this environment, parental fertility and bequest choices determine children's access to startup capital and shape children's entrepreneurial opportunities.

After solving the model, I calibrate it using data from the Panel Study of Income Dynamics (PSID). The calibration proceeds in two stages. In the first stage, I estimate individual-level income and consumption profiles over the life cycle using fixed-effects regressions, which allows me to construct expected lifetime income and consumption. I also use asset holdings at age 25, when individuals are unlikely to have accumulated substantial wealth on their own, as a proxy for intergenerational transfers. Macro-level parameters are then calibrated by Generalized Method of Moments (GMM) to match key aggregate moments, including the entrepreneurship rate and the income distribution. In the second stage, I use the estimated income profiles from the first stage to calibrate utility parameters by matching the mean of log consumption and the distribution of fertility.

The calibration suggests that the U.S. credit market is relatively well-functioning, in line with the baseline assumptions in the cross-country studies relating to financial development (Buera et al., 2011). The calibrated model also helps reconcile the evidence in Hurst and Lusardi (2004), who find that entrepreneurial entry is related to wealth only near the top of the wealth distribution. In my framework, this pattern arises because borrowing constraints

depend on the interaction between entrepreneurial talent and wealth. In a well-functioning financial market, borrowing constraints bind mainly for a few high-talent individuals with low wealth and for low-talent individuals with high wealth who can leverage their assets into entry.

In the first counterfactual analysis, reducing financial barriers, relative to the baseline model calibrated to the US, improves economic development, but at the cost of slightly reducing entrepreneurship and increasing income inequality. This occurs because more developed financial markets enable highly productive individuals to attract more capital by offering higher interest rates, thereby amplifying returns on wealth and crowding out marginal entrepreneurs. These findings underscore the importance of designing finance, tax, and transfer policies that account for both economic development and concerns about income concentration.

In the second counterfactual analysis, shutting down the endogenous fertility channel, as is common in the existing literature, can overestimate inequality and intergenerational mobility. When fertility is treated as exogenous, the model fails to capture fertility's dilution effect on family resources and its influence on wealth distribution. This result highlights the importance of considering the endogenous fertility channel in policy evaluations.

This paper contributes to three strands of the literature. First, it contributes to the literature on fertility and intergenerational resource allocation (Aso, 2024; Cooke et al., 2017; Daruich and Kozlowski, 2016; Pestieau, 1984) by emphasizing fertility's dilution effect on intergenerational monetary transfers and, thereby, its impact on children's entrepreneurial opportunities. Second, it contributes to the literature on entrepreneurship and borrowing constraints (Buera, 2008; Evans and Jovanovic, 1989) by showing that family size is an important determinant of the initial resources available for entrepreneurs to overcome borrowing constraints. Third, it contributes to the literature on entrepreneurship's influence on inequality and intergenerational mobility (Brüggemann, 2021; Cagetti and De Nardi, 2006;

Ivanova, 2023; Quadrini, 1999, 2000; Scheuer, 2014; Terajima, 2006) by highlighting how fertility-driven dilution of family resources interacts with the leverage effect of entrepreneurship to shape the persistence of income inequality across generations when financial barriers exist.

The remainder of the paper is structured as follows. Section 2 reviews the related literature and outlines the paper's contribution. Section 3 presents the benchmark model without financial barriers. Section 4 introduces borrowing constraints into the framework. Section 5 describes the data and calibration strategy. Section 6 reports the counterfactual analysis. Section 7 concludes.

2 Related Literature

The key mechanism in this paper relates to the literature on fertility-driven dilution of family resources. The classical quantity–quality trade-off literature defines "quality" primarily in terms of children's human capital investment (Becker, 1960; Docquier, 2004) and fertility choices often affect intergenerational mobility and inequality through human capital accumulation and its impact on labor income (Ehrlich and Kim, 2007; Moav, 2005). A related strand of literature emphasizes that fertility also dilutes wealth transfers among children and thereby shapes intergenerational mobility and income inequality through wealth accumulation (Aso, 2024; Cooke et al., 2017; Pestieau, 1984). Complementary work studies how monetary transfers affect wealth and income distributions without explicitly modeling fertility (De Nardi, 2004; De Nardi and Yang, 2016). More recently, scholars have combined fertility choices with both human capital investment and monetary transfers to study how family choices influence social mobility in a more comprehensive way (Daruich and Kozlowski, 2016), though abstracting from the entrepreneurship channel.

However, the influence of intergenerational wealth transmission on mobility and inequality is not limited to the direct effect of inherited wealth, as wealth can have a leverage effect through entrepreneurship. When individuals receive monetary transfers from parents, they can relax borrowing constraints and finance a larger amount of capital when starting a business. Empirically, the relationship between wealth and entrepreneurial entry is nuanced. Although Hurst and Lusardi (2004) argues that wealth holdings only influence entrepreneurial entry at the top of the wealth distribution, Fairlie and Krashinsky (2012) shows that the aggregate relationship between wealth and entry can be misleading because low-wealth individuals are more likely to enter entrepreneurship after job loss. When job losers and non-job losers are analyzed separately, entrepreneurial entry increases with wealth within each group. This evidence suggests that wealth still matters for entrepreneurship, but its effect can be obscured by differences in the reasons why individuals enter business ownership.

Entrepreneurship, in turn, is a key driver of wealth concentration and persistence at the top of the income and wealth distributions (Brüggemann, 2021; Quadrini, 1999, 2000; Scheuer, 2014; Terajima, 2006). As a result, the monetary transfers within the family across generations lead to the intergenerational transmission of inequality. Individuals from rich families are more likely to overcome entrepreneurial financial barriers and become entrepreneurs, thereby remaining at the top of income and wealth distributions through business ownership.

Existing quantitative work captures the influence of bequests on entrepreneurship and, consequently, income and wealth distributions, but typically abstracts from the endogenous fertility channel. Cagetti and De Nardi (2006) models intergenerational transmission through discount factors but does not explicitly include endogenous bequests. Ivanova (2023) examines how family monetary transfers alleviate financial barriers to entrepreneurship and shape wealth and income distributions in a dynamic model, but it omits the dilution effect of fer-

tility on family resources. However, empirical studies show that family structure and size can affect entrepreneurship (Vladasel, 2023), aligning with the dilution mechanism in family resource allocation.

The main contribution of this paper is to integrate these strands of research and show how fertility-driven dilution of family resources interacts with credit constraints to shape entrepreneurship and inequality. Fertility determines the amount of inherited resources available to each child, while credit constraints determine how much these resources matter for entrepreneurial entry, firm scale, and subsequent wealth accumulation. These mechanisms reinforce each other: smaller transfers make borrowing constraints more binding, and binding constraints can prevent talented individuals from becoming entrepreneurs, operating at an efficient scale, and accumulating wealth through entrepreneurship. Combining these mechanisms shows that fertility affects inequality not only through human capital or direct transfers, but also through entrepreneurship's leverage effect on the accumulation of income and wealth.

3 Benchmark Model

I first build a model without borrowing constraints as the benchmark. I adopt a discrete-time OLG model where people experience two stages of life: child and adult. The children consume passively. The adults make entrepreneurship choices and maximize their utility.

3.1 Utility Maximization

After knowing their income, parents choose consumption c_{it}, total bequest to all children m_{it} and the number of children n_{it} to maximize their utility:

$$\max_{\{c_{it},n_{it},m_{it}\}} \left(\theta_1 c_{it}^{1-\frac{1}{\epsilon}} + \theta_2 m_{it}^{1-\frac{1}{\epsilon}} + (1-\theta_1-\theta_2) n_{it}^{1-\frac{1}{\epsilon}}\right)^{\frac{\epsilon}{\epsilon-1}} \tag{1}$$

$$\text{Subject to:} \quad c_{it} + m_{it} = (1-\tau n_{it}) y_{it}$$

The parameter ϵ is the elasticity of substitution between consumption, total bequest, and fertility. θ_1 and θ_2 represent how much people value consumption and bequest respectively. y_{it} is individual i_t's income. τ is the time cost of having a child. To raise n_{it} children, the parent needs to pay $\tau \times n_{it}$ fraction of their income y_{it}.

The parent's optimal consumption, fertility, and bequest choices are

$$c_{it}^* = \frac{1}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it} \tag{2}$$

$$m_{it}^* = \frac{(\frac{\theta_2}{\theta_1})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it} \tag{3}$$

$$n_{it}^* = \frac{(\frac{1-\theta_1-\theta_2}{\theta_1 \tau})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it}^{1-\epsilon} \tag{4}$$

Consumption and bequest are increasing in an individual's income. The monotonicity of the number of children depends on the value of ϵ. When $\epsilon \leq 1$, n_{it} increases in y_{it}. However, if $\epsilon > 1$, n_{it} will first increase in income when $y_{it} < (\frac{1-\theta_1-\theta_2}{\theta_1 \tau})(\frac{1+(\frac{\theta_2}{\theta_1})^\epsilon}{\epsilon-1})^{\frac{1}{\epsilon}}$. As income increases above the threshold, the number of children will decrease in income due to the increased child cost. Figure 1 presents the relationship between income and the number of children for different values of ϵ.

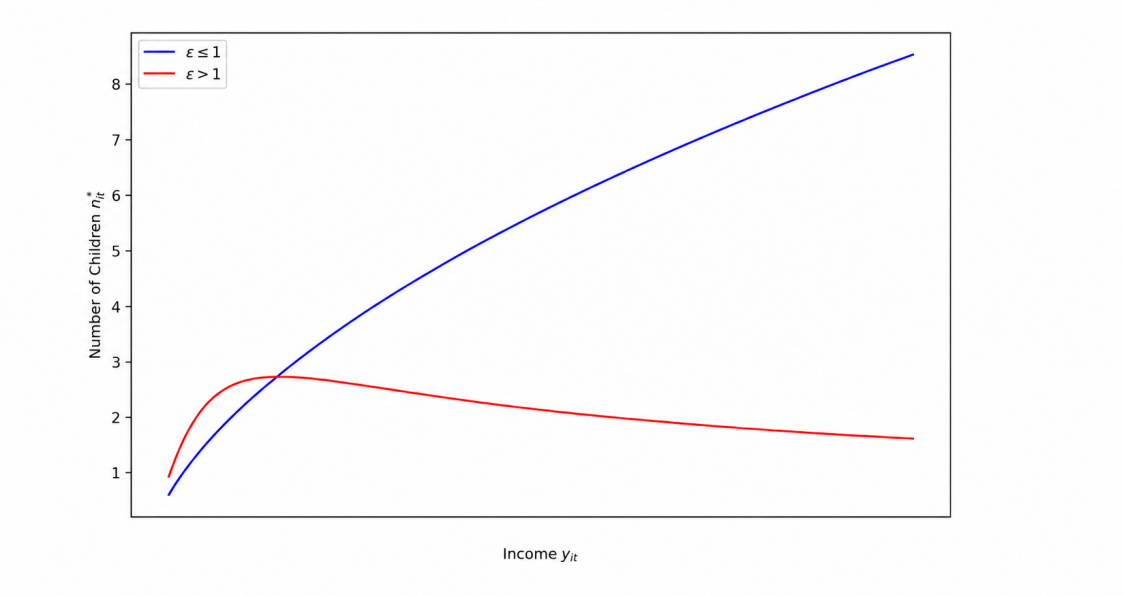


Figure 1: Relationship between Income and Number of Children

I define $b_{it} = \frac{m_{it}}{n_{it}}$ as the average bequest. Each child in the family receives the same bequest amount, b_{it}. When the child grows up and becomes an adult, they inherit the amount of b_{it} as their asset. In aggregate, for the next generation $t+1$, a fraction of $\frac{n_{it}}{\sum_{i_t}(n_{it})}$ people will receive b_{it} amount of bequests as their assets. Parents' optimal bequest and fertility choices determine the distribution of assets for the next generation.

3.2 Individuals' Setup

Children inherit parental bequests as assets:

$$a_{\text{children},t} = b_{\text{parent},t-1} \tag{5}$$

At the same time, entrepreneurial ability is given by the following equation:

$$\log(z_{\text{children},t}) = \rho \log(z_{\text{parent},t-1}) + \bar{z} + \epsilon_{it} \tag{6}$$

$\epsilon_{it} \sim N(0, \sigma_\epsilon^2)$. Each individual's talent is partially inherited from the parent and $0 < \rho < 1$. In the long run, the individual-level talent would converge to $e^{\frac{\bar{z}}{1-\rho}}$.

3.3 Occupation

After learning their asset level a_{it} and entrepreneurial talent z_{it}, the individual i_t faces an occupational choice: to become an entrepreneur or a worker. This decision depends on the income in each occupation. Specifically, if she becomes an entrepreneur, she faces the following profit maximization problem:

$$\max_{\{K_{it}, L_{it}\}} \pi_{it} = (A \times z_{it})^{1-\alpha-\beta} (K_{it})^{\alpha} L_{it}^{\beta} - R_t K_{it} - W_t L_{it} \tag{7}$$

Thus, her optimal choices if she were to become an entrepreneur are:

$$K_{it}^* = (\frac{\alpha}{R_t})^{\frac{1-\beta}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} A z_{it} \tag{8}$$

$$L_{it}^* = (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{1-\alpha}{1-\alpha-\beta}} A z_{it} \tag{9}$$

and the profit is:

$$\pi_{it}^* = A z_{it} (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} (1-\alpha-\beta) \tag{10}$$

where I assume the depreciation rate of capital is 1.

As an entrepreneur, the individual can also earn interest on the asset from the bank. Consequently, the total income of an entrepreneur should be:

$$y_{it}^{\mathrm{E}} = \pi_{it}^{*} + R_t a_{it} \tag{11}$$

On the other hand, an individual provides one unit of labor inelastically as a worker. Workers are matched with the entrepreneurs randomly, and each worker earns the average expected wage in the labor market. Thus, if she becomes a worker, her income is:

$$y_{it}^{\mathrm{W}} = W_t + R_t a_{it} \tag{12}$$

By comparing her income as an entrepreneur y_{it}^{E} and her income as a worker y_{it}^{W} , the individual makes the optimal occupational choice according to equation (13). An individual chooses to become an entrepreneur if and only if $y_{it}^{\mathrm{E}} \geq y_{it}^{\mathrm{W}}$.

$$o_{it} = \begin{cases} 1 & \text{if} \quad y_{it}^{\mathrm{E}} \geq y_{it}^{\mathrm{W}} \\ 0 & \text{if} \quad y_{it}^{\mathrm{E}} < y_{it}^{\mathrm{W}} \end{cases} \tag{13}$$

This implies that she becomes an entrepreneur if and only if $z_{it} \geq z_t^e$, where:

$$z_t^e = \frac{1}{A} \frac{\beta}{1-\alpha-\beta} \left(\frac{R_t}{\alpha}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{W_t}{\beta}\right)^{\frac{1-\alpha}{1-\alpha-\beta}} \tag{14}$$

which does not depend on individual-level characteristics. Individuals with entrepreneurial talent higher than z_{it}^e choose to become entrepreneurs with income y_{it}^E. Those below the

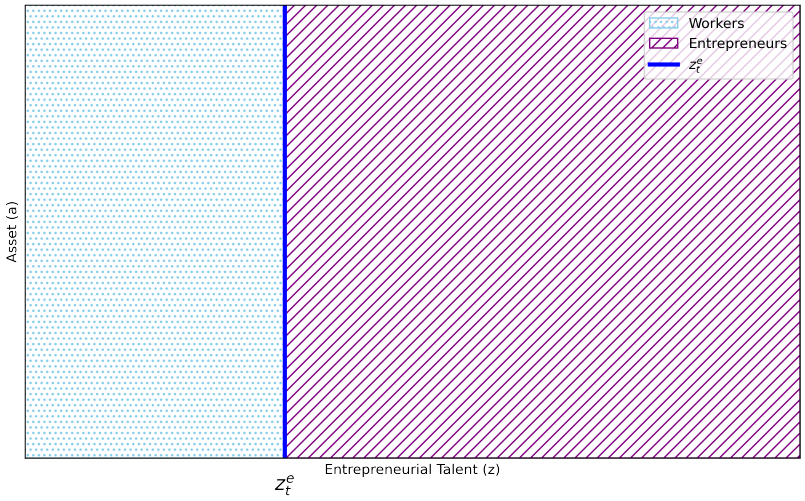


Figure 2: Occupation Choices in the Benchmark Model

threshold will become workers with income y_{it}^W. Figure 2 shows individuals' occupation choices in the benchmark model.

3.4 Equilibrium

Let $\mathbf{x} = (a, z)$ denote an individual's state, where a is inherited assets and z is entrepreneurial talent. Let $d_t(\mathbf{x})$ be the distribution of individuals over states at time t.

A competitive equilibrium is a sequence of prices $\{R_t, W_t\}_{t=0}^{\infty}$, distributions $\{d_t(\mathbf{x})\}_{t=0}^{\infty}$, and decision rules $c(\mathbf{x})$, $n(\mathbf{x})$, $m(\mathbf{x})$, $o(\mathbf{x})$, $K(\mathbf{x})$, $L(\mathbf{x})$, and $y(\mathbf{x})$ such that, in each period t, the following conditions hold:

(a) Adults choose $c(\mathbf{x})$, $n(\mathbf{x})$, and $m(\mathbf{x})$ to maximize utility in (1).

(b) Entrepreneurs choose $K(\mathbf{x})$ and $L(\mathbf{x})$ to maximize profits in (7).

(c) Individuals choose occupations according to (13). Their income $y(\mathbf{x})$ is given by the corresponding worker or entrepreneur income.

(d) Labor and capital markets clear:

$$\int_{\{\mathbf{x}:o(\mathbf{x})=0\}} d_t(\mathbf{x}) = \int_{\{\mathbf{x}:o(\mathbf{x})=1\}} L(\mathbf{x}) d_t(\mathbf{x}), \tag{15}$$

$$\int a\, d_t(\mathbf{x}) = \int_{\{\mathbf{x}:o(\mathbf{x})=1\}} K(\mathbf{x}) d_t(\mathbf{x}). \tag{16}$$

(e) The distribution evolves according to

$$d_{t+1}(\mathbf{x}') = \frac{\int n(\mathbf{x}) p(\mathbf{x}'|\mathbf{x}) d_t(\mathbf{x})}{\int n(\mathbf{x}) d_t(\mathbf{x})}. \tag{17}$$

A stationary equilibrium is a competitive equilibrium in which prices and the distribution are time invariant. That is, $R_t = R$, $W_t = W$, and $d_t = d^*$ for all t. The stationary distribution satisfies

$$d^*(\mathbf{x}') = \frac{\int n(\mathbf{x}) p(\mathbf{x}'|\mathbf{x}) d^*(\mathbf{x})}{\int n(\mathbf{x}) d^*(\mathbf{x})}. \tag{18}$$

4 Model with Financial Barriers

The structure of the model with borrowing constraints is similar to the benchmark model. Parents solve the utility-maximization problem in (1) by choosing optimal consumption, bequest, and fertility. When children grow up, they inherit their parents' bequest as the asset, following equation (5), and acquire entrepreneurial talent, as specified in equation (6). The only difference is that now entrepreneurs can either default on the contract or repay the bank. The bank will set a limit on the capital it lends to each individual to prevent defaults.

4.1 Bank and Credit Market

After learning about their asset and entrepreneurial talent, individuals go to the bank and assess their financial situation. On the one hand, individuals save their money in the bank to earn interest. On the other hand, they can borrow money from the bank to start a firm. However, the credit market is not perfect due to the imperfect enforceability of contracts. Following Buera et al. (2011), I assume that individuals can renege on their contract and keep a $1-s$ fraction of revenue net of labor cost. The only punishment is that they will lose the assets deposited in the bank.

The bank earns zero profit. The interest rate the bank pays is the same as the interest rate the bank lends. To further simplify the model, I assume people put all their assets in the bank and borrow the capital to start a firm. Under the model's assumption, this is equivalent to the case that adults use their assets to start a firm and save the rest in the bank if they have excess assets or borrow from the bank if they do not have sufficient capital.

If $s \geq \frac{\alpha}{1-\beta}$, when the bank has strong seizure power, the economy does not have any financial barriers, and the credit market is perfect, we are back to the benchmark model. When $0 \leq s < \frac{\alpha}{1-\beta}$, being aware that individuals can default, the bank will lend up to an upper capital limit K_{it}^{limit} to prevent default. The bank wants to ensure that when utilized capital is smaller than or equal to K_{it}^{limit}, default behavior is less profitable than non-default behavior. Mathematically, the bank sets up K_{it}^{limit} for each individual i_t such that:

$$\begin{aligned} &\max_{\{K_{it}\in[0,K_{it}^{limit}],L_{it}\}} (1-s)\{(Az_{it})^{1-\alpha-\beta}(K_{it})^{\alpha}L_{it}^{\beta} - W_t L_{it}\} - a_{it} \\ &\leq \max_{\{K_{it}\in[0,K_{it}^{limit}],L_{it}\}} \{(Az_{it})^{1-\alpha-\beta}(K_{it})^{\alpha}L_{it}^{\beta} - R_t K_{it} - W_t L_{it}\} + R_t a_{it} \end{aligned} \tag{19}$$

To make it easier to compare, I define $\tilde{s} \in [0,1]$ where $s = \tilde{s} \times \frac{\alpha}{1-\beta}$.

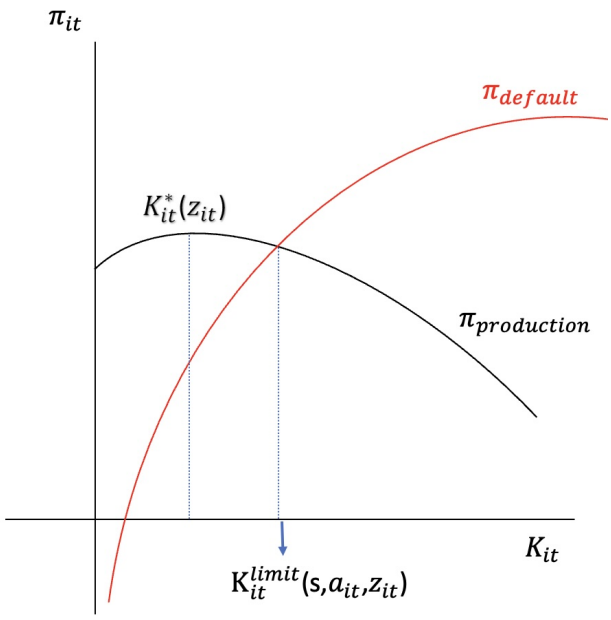


Figure 3: Capital Limit for Unbounded Individual

Figure 3 and Figure 4 illustrate the capital limit for unbounded and bounded individuals separately. In both figures, the bank wants to set K_{it}^{limit} so that the default profit is no greater than the production profit. Accordingly, the highest level of K_{it}^{limit} would be given by the intersection of π_{default} and $\pi_{\text{production}}$. In figure 3, the capital limit that the bank is willing to lend is higher than the individual's optimal capital level, so this individual will not be bound by the bank, and they can produce at the optimal capital level. In this case, I assume that the highest amount of capital the bank will lend would be K_{it}^* to simplify the model.

However, in Figure 4, $K_{it}^{limit} < K_{it}^*$. In this case, the capital individuals can borrow is less than the optimal capital level. They can only produce at a suboptimal capital level if they become entrepreneurs. Individuals are unbounded if and only if:

$$\frac{a_{it}}{z_{it}} \geq A \times \frac{\alpha - s(1-\beta)}{1+R_t} \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{W_t}\right)^{\frac{\beta}{1-\alpha-\beta}} \tag{20}$$

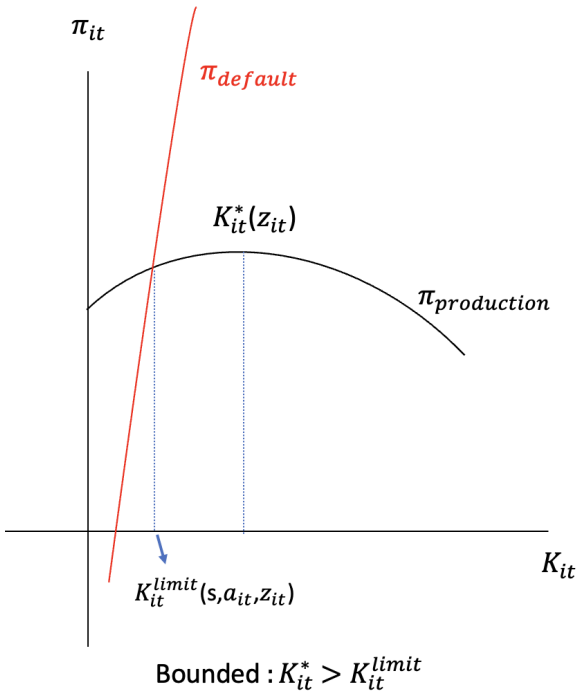


Figure 4: Capital Limit for Bounded Individual

In equilibrium, no one will default. Moreover, K_{it}^{limit} is a function of the bank's seizure rate s, personal assets a_{it} and entrepreneurial ability z_{it}. $K_{it}^{limit}(s, a_{it}, z_{it})$ is increasing in s, a_{it} and z_{it} and concave in a_{it} and z_{it}.

4.2 Occupation, Income, and Utility

Individuals choose their occupations given their capital limit K_{it}^{limit} determined by the bank. They first face the following production problem with capital limit K_{it}^{limit}

$$\max_{\{K_{it} \leq K_{it}^{limit}, L_{it}\}} \pi_{it} = (Az_{it})^{1-\alpha-\beta}(K_{it})^{\alpha} L_{it}^{\beta} - R_t K_{it} - W_t L_{it} \tag{21}$$

For unbounded individuals, they will produce at the optimal capital level K_{it}^* based on equation (8) and earn profit π_{it}^* given in equation (10). For bounded individuals, they can

only produce at a suboptimal level K_{it}^{limit}. Based on (19), an individual's capital limit K_{it}^{limit} is given implicitly by:

$$s(1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}}(K_{it}^{limit})^{\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} - R_t K_{it}^{limit} + (1+R_t)a_{it} = 0 \tag{22}$$

The profit for bounded entrepreneurs is:

$$\pi_{it}^{B} = (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} K_{it}^{limit\,\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}(1-\beta) - R_t K_{it}^{limit} \tag{23}$$

As in the benchmark model, entrepreneurs can earn interest on their assets from the bank in addition to production profits. As a result, the total income of an entrepreneur would be:

$$y_{it}^{\mathrm{E}} = \begin{cases} y_{it}^{\mathrm{UE}} = \pi_{it}^{*} + R_t a_{it} & \text{if unbounded} \\ y_{it}^{\mathrm{BE}} = \pi_{it}^{B} + R_t a_{it} & \text{if bounded} \end{cases} \tag{24}$$

The worker's income is the same as the benchmark model given by equation (12). Individuals choose the occupation that earns the highest income following equation (13).

For unbounded individuals, the entrepreneurial cutoff is the same as the benchmark model given by equation (14). For bounded individuals, their entrepreneurial cutoff will be given by equation (25). Bounded individuals with entrepreneurial ability higher than or equal to z_{it}^{BE} become bounded entrepreneurs.

$$z_{it}^{BE} = \frac{1}{A}\left(\frac{W_t + R_t K_{it}^{limit}}{(1-\beta)(K_{it}^{limit})^{\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}}\right)^{\frac{1-\beta}{1-\alpha-\beta}} \tag{25}$$

Figure 5 shows individuals' occupational choices when financial barriers exist. The Y-axis

shows the asset level, while the X-axis presents the entrepreneurial talent. The orange line indicates the bound status given by equation (20). Individuals above the line are unbounded, while those under the line are bounded. The vertical blue line is the entrepreneurial cutoff for unbounded individuals given in equation (14). Unbounded individuals with talent equal to or higher than the cutoff will become entrepreneurs. The green curve demonstrates the entrepreneurial cutoff for bounded people according to equation (25). Bounded people above the line choose to become bounded entrepreneurs.

One comparative static analysis we could do here is to check the influence of the bank's seizure rate s while fixing R and W. In a model without financial barriers, individuals with lower entrepreneurial ability choose to become workers because the labor income is higher than entrepreneurial income: $y_{it}^{W} > y_{it}^{UE}$. When we add borrowing constraints, we know that bounded entrepreneurial income must be no greater than unbounded entrepreneurial income: $y_{it}^{UE} \geq y_{it}^{BE}$. As a result, we should have $y_{it}^{W} > y_{it}^{UE} \geq y_{it}^{BE}$. Individuals to the left of the unbounded entrepreneurial cutoff still choose to become workers in a world with financial barriers. For those who are entrepreneurs in a world without financial barriers, there are three situations. Those with low assets now become workers since $y_{it}^{UE} \geq y_{it}^{W} > y_{it}^{BE}$. Individuals with intermediate asset levels now become bounded entrepreneurs because $y_{it}^{UE} \geq y_{it}^{BE} > y_{it}^{W}$. Finally, individuals with the highest level of assets remain unbounded entrepreneurs. Comparing the benchmark model with the model considering borrowing constraints, we can see occupation distortions and mismatches between entrepreneurial talent and capital.

Given their occupation and corresponding income, an individual maximizes their utility according to (1). The optimal decision rules are precisely the same as those in the benchmark model listed in equation (2), equation (3), and equation (4).

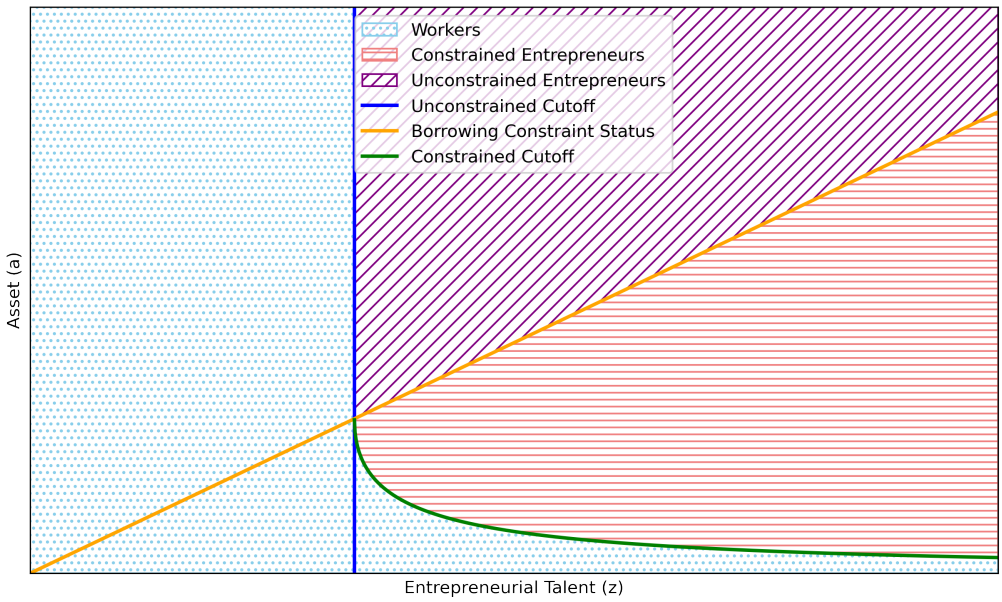


Figure 5: Occupation Choices in the Model with Financial Barriers

4.3 Equilibrium

Let $\mathbf{x} = (a, z)$ denote an individual's state, where a is inherited assets and z is entrepreneurial talent. Let $d_t(\mathbf{x})$ be the distribution of individuals over states at time t. Given the bank's seizure rate s, the transition function $p(\mathbf{x}'|\mathbf{x})$ governs the evolution of individual states across generations.

A competitive equilibrium is a sequence of prices $\{R_t, W_t\}_{t=0}^{\infty}$, distributions $\{d_t(\mathbf{x})\}_{t=0}^{\infty}$, and decision rules $c(\mathbf{x})$, $n(\mathbf{x})$, $m(\mathbf{x})$, $o(\mathbf{x})$, $B(\mathbf{x})$, $K^{limit}(\mathbf{x})$, $K(\mathbf{x})$, $L(\mathbf{x})$, and $y(\mathbf{x})$ such that, in each period t, the following conditions hold:

(a) Adults choose $c(\mathbf{x})$, $n(\mathbf{x})$, and $m(\mathbf{x})$ to maximize utility in (1).
(b) The bank sets the capital limit $K^{limit}(\mathbf{x})$ for each individual according to (19), which determines the individual's bound status $B(\mathbf{x})$. Let $B(\mathbf{x}) = 1$ if the individual is constrained, and $B(\mathbf{x}) = 0$ if the individual is unconstrained.

(c) Entrepreneurs maximize profits subject to $K(\mathbf{x}) \leq K^{limit}(\mathbf{x})$ in (21).

(d) Individuals choose occupations according to (13). Their income $y(\mathbf{x})$ is given by the corresponding worker, unconstrained entrepreneur, or constrained entrepreneur income.

(e) Labor and capital markets clear:

$$\int_{\{\mathbf{x}:o(\mathbf{x})=0\}} d_t(\mathbf{x}) = \int_{\{\mathbf{x}:o(\mathbf{x})=1,\, B(\mathbf{x})=0\}} L^*(\mathbf{x}) d_t(\mathbf{x}) + \int_{\{\mathbf{x}:o(\mathbf{x})=1,\, B(\mathbf{x})=1\}} L^{limit}(\mathbf{x}) d_t(\mathbf{x}). \tag{26}$$

$$\int a\, d_t(\mathbf{x}) = \int_{\{\mathbf{x}:o(\mathbf{x})=1,\, B(\mathbf{x})=0\}} K^*(\mathbf{x}) d_t(\mathbf{x}) + \int_{\{\mathbf{x}:o(\mathbf{x})=1,\, B(\mathbf{x})=1\}} K^{limit}(\mathbf{x}) d_t(\mathbf{x}). \tag{27}$$

(f) The distribution evolves according to

$$d_{t+1}(\mathbf{x}') = \frac{\int n(\mathbf{x}) p(\mathbf{x}'|\mathbf{x}) d_t(\mathbf{x})}{\int n(\mathbf{x}) d_t(\mathbf{x})}. \tag{28}$$

A stationary equilibrium is a competitive equilibrium in which prices and the distribution are time invariant. That is, $R_t = R$, $W_t = W$, and $d_t = d^*$ for all t. The stationary distribution satisfies

$$d^*(\mathbf{x}') = \frac{\int n(\mathbf{x}) p(\mathbf{x}'|\mathbf{x}) d^*(\mathbf{x})}{\int n(\mathbf{x}) d^*(\mathbf{x})}. \tag{29}$$

5 Calibration

In this section, I use data from the Panel Study of Income Dynamics (PSID) to calibrate the model's parameters when financial barriers exist. I will first describe the calibration strategy and then show details about the calibration method and results.

5.1 Calibration Strategy Overview

The inputs for the calibration are the asset and talent distributions estimated from PSID. After estimating the talent and asset distributions, I first calibrate four macro-level parameters: the TFP shifter A, the capital share α, the labor share β, and the seizure rate s. The targeted moments are the entrepreneurship rate, the top 5% income share, the log mean of income, and the log median of income.

After calibrating the macro-level parameters, I use the estimated parameter values and the distribution of talent and assets to generate the estimated income. Then I calibrate the utility parameters using the estimated income to target the log mean of consumption and the distribution of fertility. The estimated utility parameters include consumption weight θ_1, bequest weight θ_2, elasticity ϵ and cost per child τ.

5.2 PSID Data

The PSID is longitudinal data from a representative sample of U.S. households. It includes intergenerational data on income, consumption, fertility, and occupation over a long period. Before 1999, the survey was conducted annually. After the 1999 wave, it switched to a biennial cycle. My sample consists of the 1968 to 2011 waves of the PSID, and I keep observations for household heads. However, the consumption data regarding detailed categories is not available until the 1999 wave. To extend the dataset, I use consumption data from Attanasio and Pistaferri (2014), which uses socioeconomic variables, prices, and food consumption data consistently available in the PSID to impute consumption before 1999.

One concern is that although PSID provides income and consumption data at the annual or biennial level, some years may be missing. It may not contain an individual's income and consumption data throughout their life cycle. Therefore, I estimate the income and

consumption data for one particular year using the following equation. For y_{it} and c_{it}:

$$\ln y_{it} = \beta_0 + \beta_1 \times age_{it} + \omega_i^y \tag{30}$$

$$\ln c_{it} = \gamma_0 + \gamma_1 \times age_{it} + \omega_i^c \tag{31}$$

t indicates the year when the individuals are aged between 25 and 65. ω_i^y and ω_i^c are the individual fixed effects for income and consumption, which also capture cohort fixed effects. After running the regression, I store the individual fixed effects and estimate yearly income and consumption using the regression coefficients. To get the lifetime income, I use a constant interest rate $R = 5\%$ to discount all the yearly incomes from age 25 to age 65. I use the same method to calculate the expected lifetime consumption.

Figure 6 shows the estimated and real log income levels. Similarly, Figure 7 shows the relationship between the estimated and real log consumption.

In my model, the asset level refers to the bequest inherited from the parent. Since the bequest data from the PSID contains a lot of missing values, I define the inherited asset as the asset level at age 25 and estimate it using the following equation:

$$a_{it} = \sigma_0 + \sigma_1 \times \text{age} + \omega_i^a \tag{32}$$

ω_i^a indicates the individual fixed effect, which also captures the cohort fixed effect. Figure 8 shows the relationship between estimated and real log asset levels. After calculating the lifetime income, lifetime consumption, and asset level at age 25, I discount all the values back to the year 1980 using the interest rate $R = 5\%$.

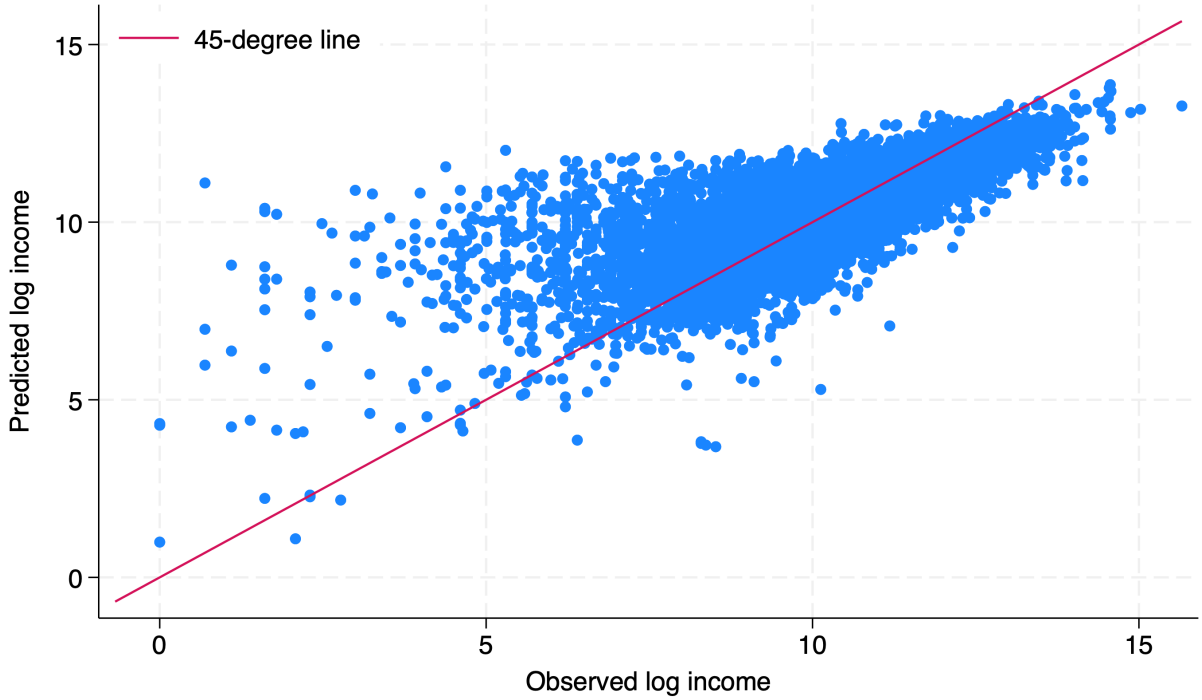


Figure 6: Real log income and estimated log income

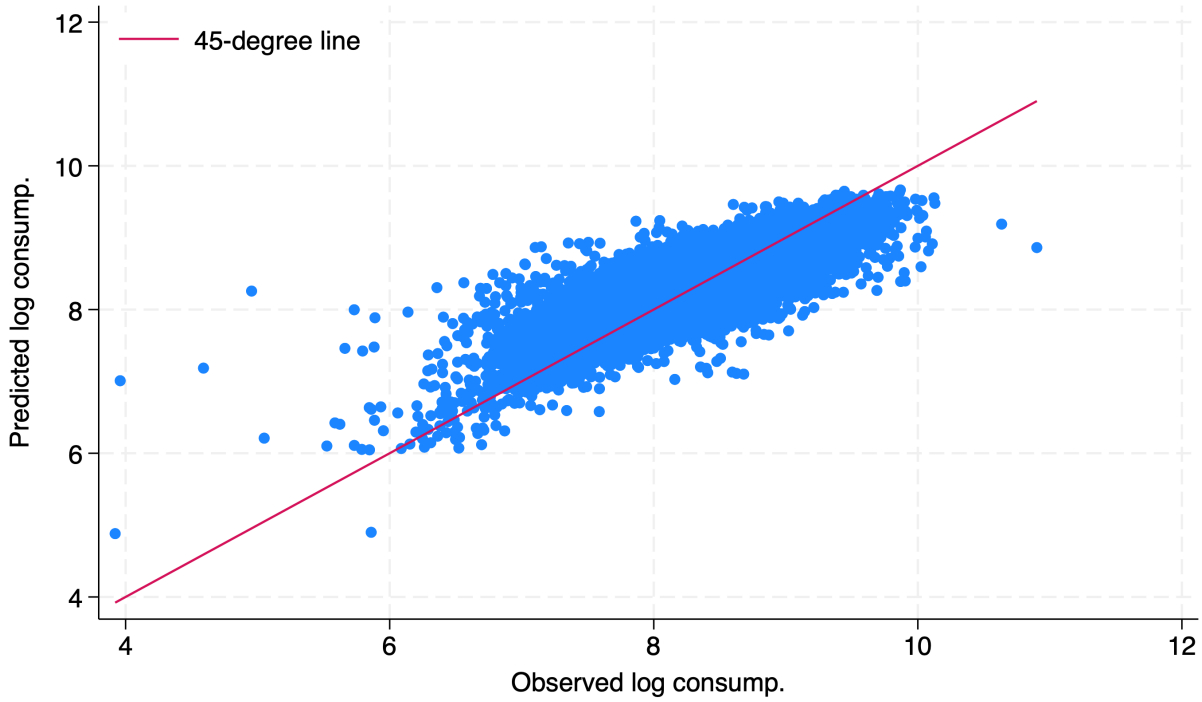


Figure 7: Real log consumption and estimated log consumption.

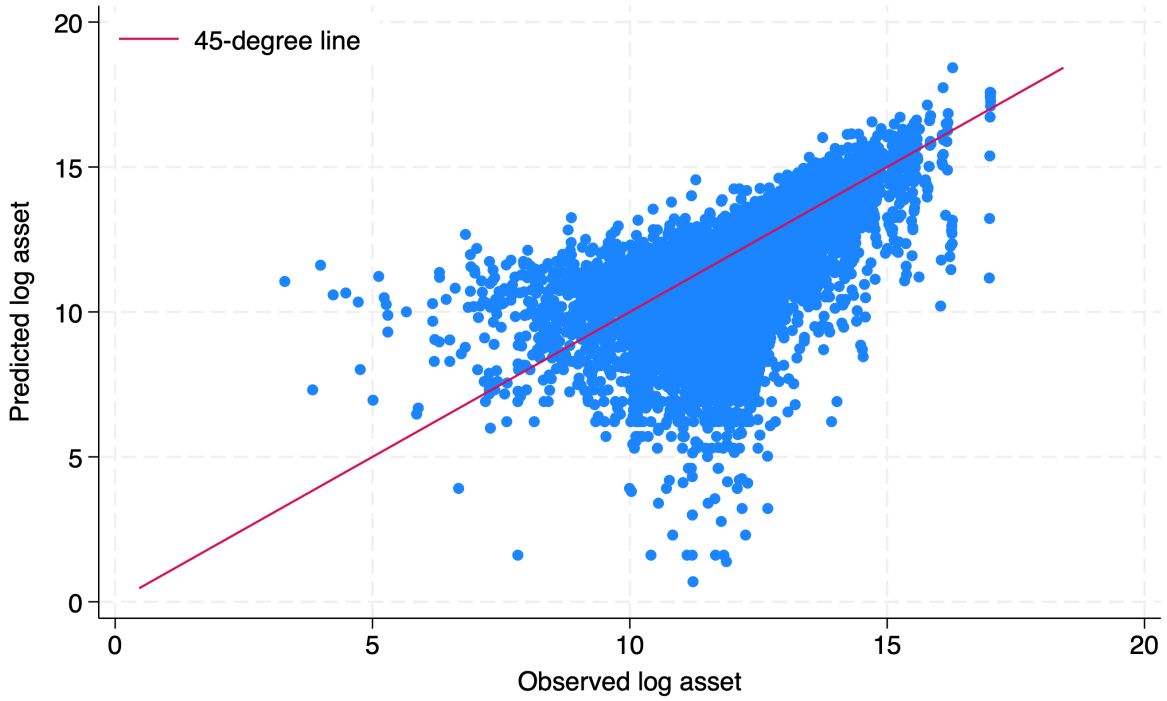


Figure 8: Real log asset and estimated log asset

Another important point to discuss is how to identify entrepreneurs in the dataset. I use two variables in the PSID dataset. The first variable relates to whether the household head is reported as self-employed. This question could capture whether the individual is working for someone else or themselves. The second variable regards whether the family owns a business or has a financial interest in any business enterprise. This question provides information about the ownership of a business. If the answers to both questions are yes, I define the individual as an entrepreneur in that particular year. I classify an individual as an entrepreneur if they are identified as an entrepreneur in at least two consecutive observed waves and in at least half of all waves in which they are observed.

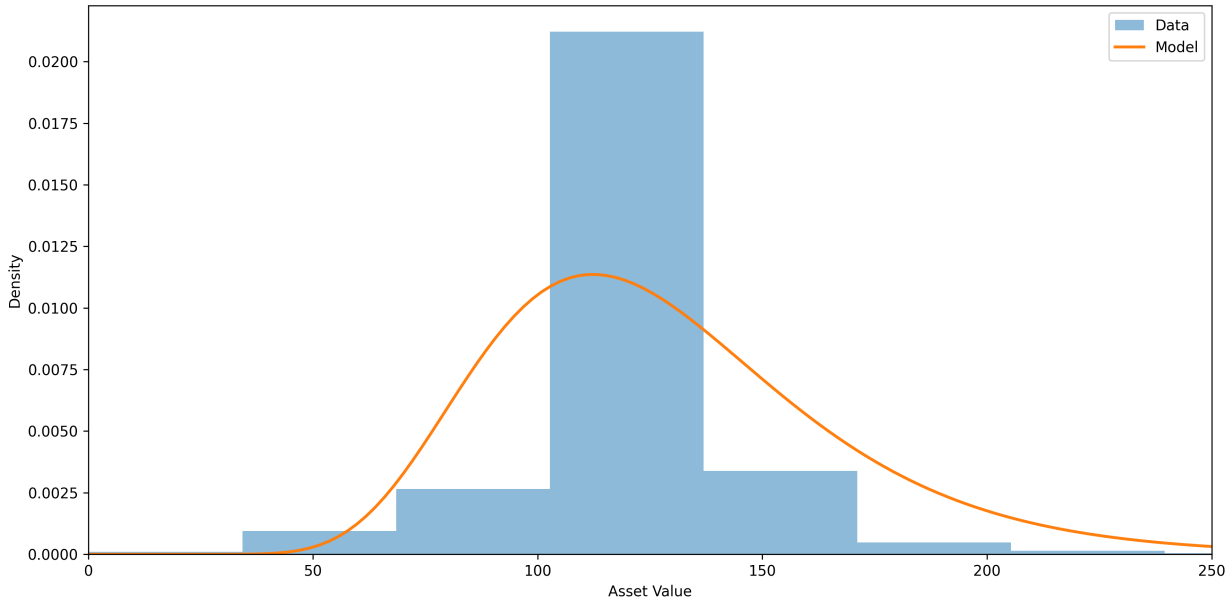


Figure 9: Asset Distribution (in units of $100,000)

5.3 Asset and Talent Distributions

Based on the literature, I assume the asset follows a lognormal distribution. The asset distribution is based on the estimated asset level at age 25. Figure 9 shows the estimated and real asset distributions.

As for the talent distribution, I first use the following equation to obtain the residuals of lifetime income as the talent measure:

$$\begin{aligned}\ln(\mathrm{y}_i) \; &= \omega_0 + \boldsymbol{\omega}_1^\top \mathbf{asset_quantile}_i + \boldsymbol{\omega}_2^\top \mathbf{race}_i + \boldsymbol{\omega}_3^\top \mathbf{disability}_i \\ &\quad + \boldsymbol{\omega}_4^\top \mathbf{gender}_i + \boldsymbol{\omega}_5^\top \mathbf{birthyear}_i \\ &\quad + \boldsymbol{\omega}_6^\top (\mathbf{asset_quantile}_i \odot \mathbf{race}_i) + \boldsymbol{\omega}_7^\top (\mathbf{asset_quantile}_i \odot \mathbf{disability}_i) \\ &\quad + \boldsymbol{\omega}_8^\top (\mathbf{asset_quantile}_i \odot \mathbf{gender}_i) + \boldsymbol{\omega}_9^\top (\mathbf{asset_quantile}_i \odot \mathbf{birthyear}_i) + \varepsilon_i \end{aligned} \tag{33}$$

In the equation, $\mathbf{asset_quantile}_i$ indicates the wealth quantile of the individual in the

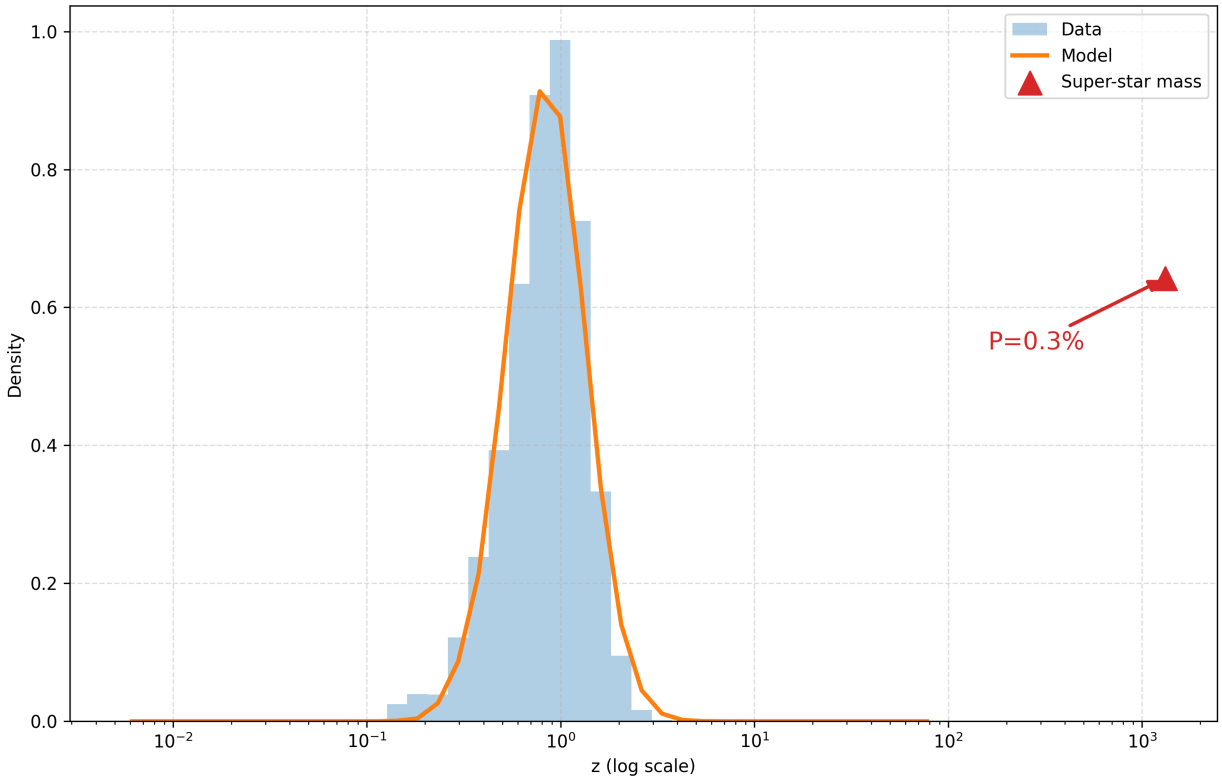


Figure 10: Talent distribution

society, and $\textbf{disability}_i$ indicates whether the individual has a disability. I treat the residual as the log talent.

I need to estimate both the talent distribution and the AR(1) process parameters. For the initial distribution, I use the log-normal distribution to simulate, but assume there exists a small fraction of super-talented individuals on the far right end. Figure 10 shows the comparison between the real data and simulated talent distribution.

I assume the intergenerational talent transmission follows the AR(1) process. Since the dataset includes only observations for the household head, and most households are male-headed, I link fathers to their children and estimate the regression in equation (34). The dataset contains 746 matched father–child pairs.

$$\ln z_{it+1} = \overline{z} + \rho \ln z_{it} + \epsilon_{it} \tag{34}$$

After estimating equation (34), I obtain the parameters reported in Table 1.

Table 1: Estimated Parameters for Talent Inheritance

Parameter	Description	Value
ρ	Persistence of talent	0.29
$\overline{z}$	Mean of log-talent	-0.1
$\delta_{\epsilon_{it}}$	Std. dev. of talent shock	0.43

Notes: This table reports the estimated parameters governing the stochastic process of talent inheritance. ρ represents intergenerational persistence, $\overline{z}$ is the mean of log talent, and $\delta_{\epsilon_{it}}$ is the standard deviation of idiosyncratic talent shocks.

5.4 Macro Parameters

After estimating the distribution of assets and talent, I have four parameters to be estimated at the macro level: the TFP shifter A, the capital share α, the labor share β, and the bank's seizure rate s. The four main targets are the fraction of entrepreneurs within the U.S., the top 5% income share, the log mean of income, and the log median of income.

5.5 Utility Parameters

After calibrating the macro-level parameters, I use the estimated income to calibrate the following four parameters in the utility maximization problem: the weight of consumption θ_1, the weight of total bequest θ_2, the elasticity of substitution ϵ, and the cost per child τ.

From the solution to the utility maximization problem in (1), we know the optimal

decisions for consumption and fertility. I use GMM (Generalized Method of Moments) to target the following moments: the log mean of consumption and the distribution of the number of children.

5.6 Calibration Results

Table 2 shows the calibrated parameter values. For the macro-level parameters, the standardized bank seizure rate $\tilde{s}$ is very close to 1, indicating that the US financial market is near perfect. The result shows that most entrepreneurs in the US are not bound by the bank unless they have high entrepreneurial talent and need to open very large firms, or they have a large amount of wealth but low entrepreneurial talent.

As for the utility parameters, the consumption weight θ_1 is around 0.13 and the total bequest weight θ_2 is around 0.22. We can observe that the elasticity $\epsilon = 4.49$, which is larger than 1, indicating that the number of children and the income level exhibit a hump-shaped relationship. The time cost of raising one child is 0.19.

Table 3 Panel A compares the model-generated moments with their empirical counterparts for the targeted moments. The model matches several key moments reasonably well, including the entrepreneurship rate and the income moments. At the same time, it understates log mean consumption. Figure 12 suggests that this gap in the mean does not come from a broad mismatch in the shape of the consumption distribution. The model captures the main distribution of consumption in the data, although it places slightly more mass at lower consumption levels and does not fully match the upper tail. Thus, the model underpredicts the average level of consumption while still matching the overall distribution reasonably well.

Figure 11 compares the income distribution generated by the model with that in the data.

Table 2: Calibrated Parameters

Parameter	Description	Value
Panel A: Financial Barriers		
$\tilde{s}$	Standardized seizure rate	0.98
Panel B: Production		
A	TFP shifter	136
α	Capital share	0.62
β	Labor share	0.28
Panel C: Preferences		
θ_1	Consumption weight	0.13
θ_2	Bequest weight	0.22
ϵ	Elasticity of substitution	4.49
τ	Per-child time cost	0.19

Notes: This table reports the calibrated parameters for the model when financial barriers exist. Panel A indicates the financial barrier parameter. Panel B shows the parameters in the production function. Panel C presents the parameters in the utility maximization problem.

Table 3: Model Fit: Targeted and Untargeted Moments

Moment	Data	Model
Panel A: Targeted Moments		
Entrepreneurship rate	0.099	0.097
Top 5% share	0.14	0.12
Log mean income	3.55	3.37
Log median income	3.43	3.29
Log mean consumption	0.85	0.62
Panel B: Untargeted Moments (Validation)		
Entrep. rate: Low-income	0.057	0.053
Entrep. rate: Middle-income	0.073	0.088
Entrep. rate: High-income	0.168	0.15
Entrep./Worker income ratio	1.469	1.661

Notes: This table compares the moments in the data with those generated by the model. Panel A reports the moments used in the calibration. Panel B reports untargeted moments. The model also targets the distribution of n. See the comparison of fertility distribution in Figure 13.

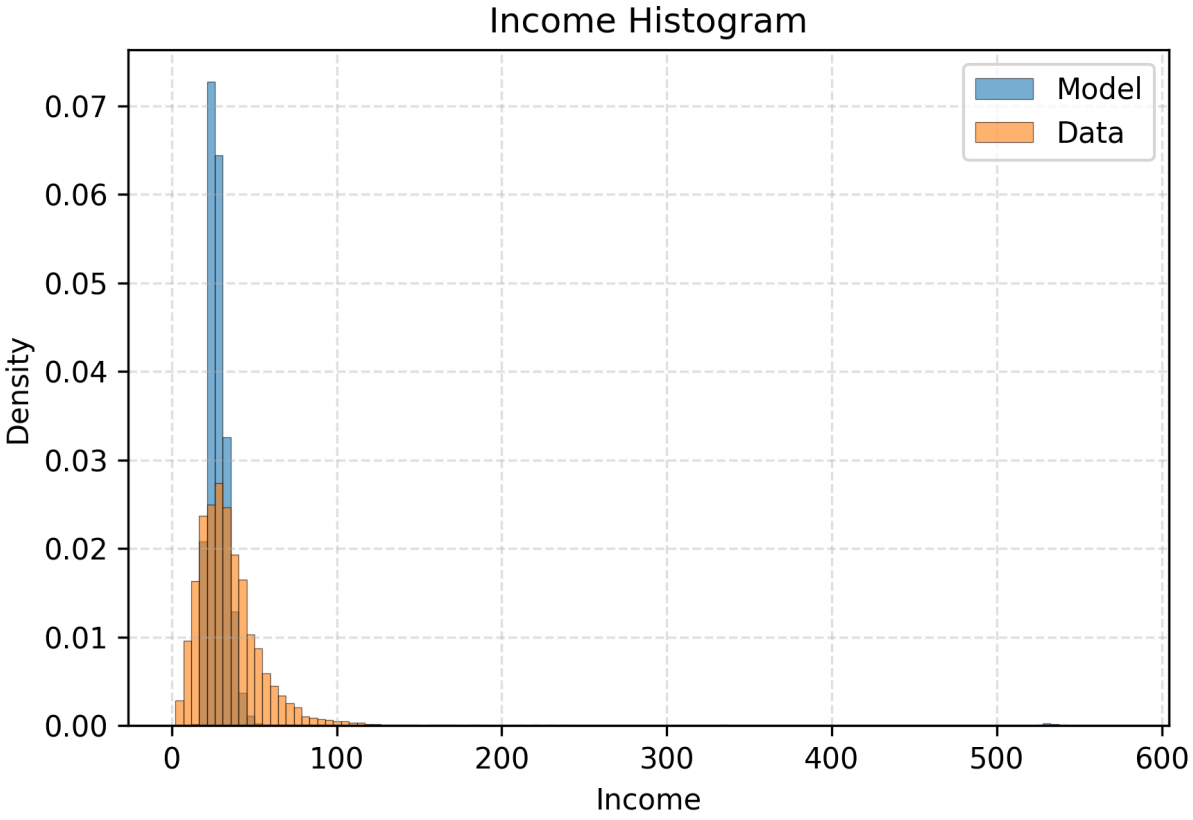


Figure 11: Income: Data vs. Model

Figures 12 and 13 report the corresponding comparisons for consumption and the number of children, respectively. The model closely matches the distributions of income and fertility. For consumption, although only log mean consumption is targeted in the calibration, the model captures the main shape of the empirical distribution reasonably well. This suggests that the lower predicted mean consumption in Table 3 reflects a level discrepancy rather than a broad mismatch in the distribution.

Finally, Table 3, Panel B, compares the model's predictions with empirical data for untargeted moments. The model captures the increasing pattern of entrepreneurial entry across income groups and generates broadly similar entrepreneurship rates. However, it overpredicts entry for the middle-income group and overstates the entrepreneur-worker income ratio.

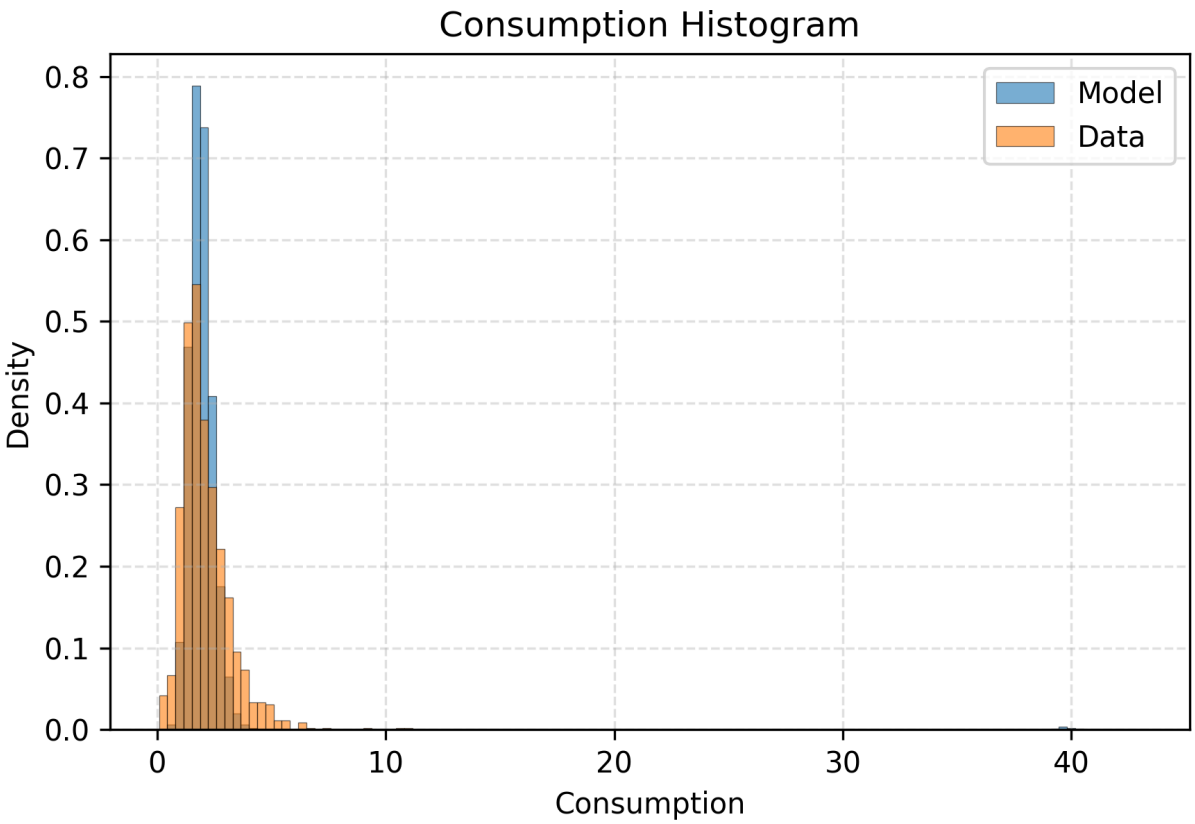


Figure 12: Consumption: Data vs. Model

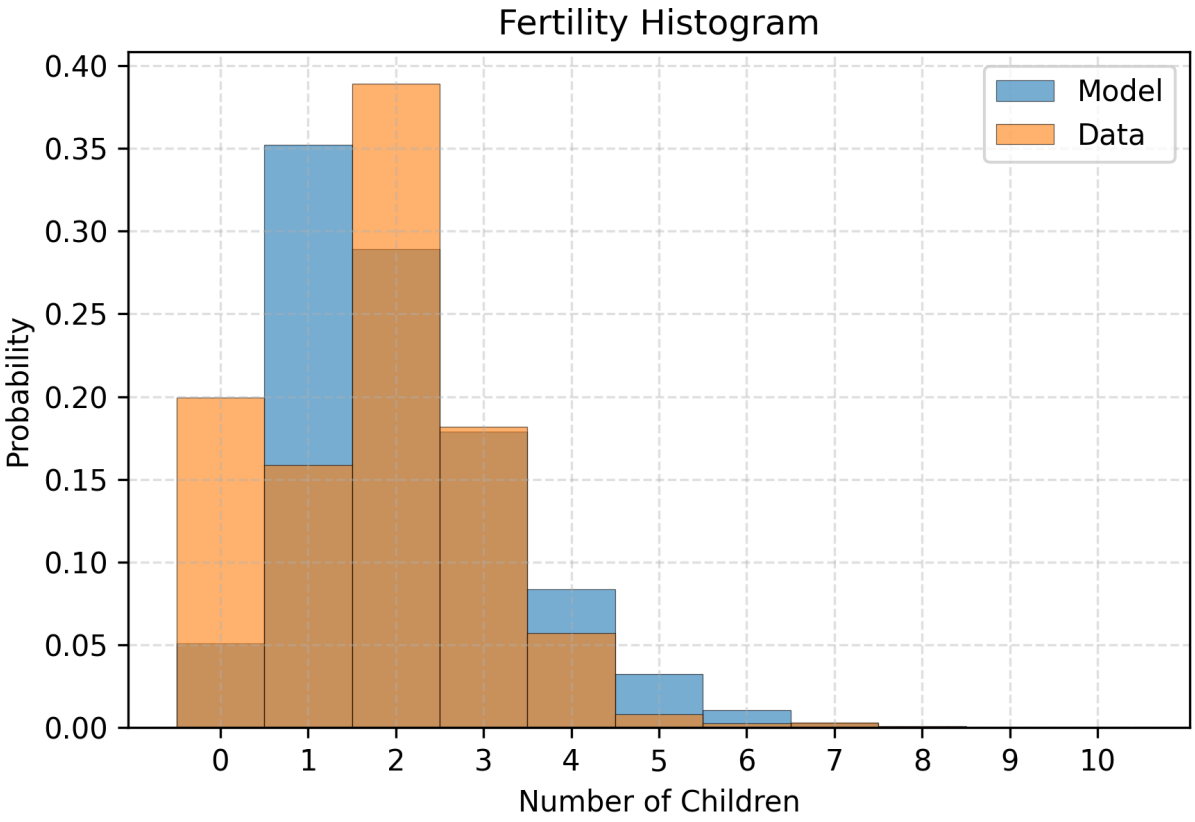


Figure 13: Fertility: Data vs. Model (Model predictions rounded to integers)

6 Counterfactual Analysis

This section presents two counterfactual exercises in the steady state. The first changes the degree of financial barriers relative to the baseline economy. I consider both a reduction in financial barriers, $\tilde{s} = 1$, and a tightening of financial barriers, $\tilde{s} = 0$. The second exercise shuts down the fertility channel by fixing fertility at $n = 1$. Comparing this fixed-fertility economy with the endogenous-fertility economy shows how fertility decisions affect the influence of financial barriers on economic development and inequality.

Table 4 reports the results of both exercises in a single table. The first three columns show the economies with endogenous fertility under the baseline, $\tilde{s} = 1$, and $\tilde{s} = 0$. These columns are used to evaluate the effects of changing financial barriers. The last three columns repeat the same financial-barrier experiments while fixing fertility at $n = 1$. Comparing the two sets of columns isolates the role of the fertility channel.

6.1 Changing Financial Barriers

I first focus on the endogenous-fertility economies in the first three columns of Table 4. Both reducing and tightening financial barriers lower the steady-state entrepreneurship rate relative to the baseline. When financial barriers are reduced to $\tilde{s} = 1$, the entrepreneurship rate falls slightly from 0.216 to 0.200. When financial barriers are tightened to $\tilde{s} = 0$, the entrepreneurship rate falls more sharply to 0.114. The entrepreneurship rate in Table 4 refers to the stationary steady state of the calibrated economy, while Table 3 reports calibration moments in the short run.

The effects on inequality are non-monotonic with respect to financial barriers. Both the Gini coefficient and the top 5% income share increase as financial barriers decline relative to the baseline. The Gini coefficient rises from 0.560 in the baseline to 0.609 under $\tilde{s} = 1$ and

to 0.643 under $\tilde{s} = 0$. The top 5% income share follows the same pattern. Intergenerational mobility, which is defined as the probability that a child's income quantile is higher than that of the parent, also declines in both counterfactual economies, falling from 0.119 in the baseline to 0.115 under $\tilde{s} = 1$ and to 0.0009 under $\tilde{s} = 0$.

Aggregate outcomes vary depending on the direction of the change in the financial barriers. Reducing financial barriers raises income per capita, consumption, and utility. Tightening financial barriers lowers all three. Reducing financial barriers raises fertility from 0.299 to 0.345, while tightening them lowers fertility to 0.274.

Table 4: Counterfactual Analysis: Financial Barriers and Fertility

	Endogenous Fertility			Fixed Fertility ($n = 1$)		
Moment	s^{base}	$\tilde{s} = 1$	$\tilde{s} = 0$	s^{base}	$\tilde{s} = 1$	$\tilde{s} = 0$
Panel A: Macroeconomic Aggregates						
Entrepreneurship rate	0.216	0.200	0.114	0.218	0.214	0.170
Income per capita	0.457	0.523	0.391	0.962	0.976	0.267
Consumption	0.012	0.013	0.009	0.043	0.044	0.012
Panel B: Inequality and Mobility						
Gini coefficient	0.560	0.609	0.643	0.623	0.627	0.634
Top 5% income share	0.555	0.594	0.630	0.632	0.637	0.641
Intergen. mobility	0.119	0.115	0.0009	0.167	0.152	0.0002
Panel C: Welfare and Demographics						
Utility	0.204	0.236	0.184	0.686	0.687	0.625
Fertility (n)	0.299	0.345	0.274	–	–	–

Notes: This table summarizes the counterfactual analysis across different levels of financial barriers ($\tilde{s}$) and fertility setups. The first three columns report results under the model with endogenous fertility. The last three columns report results for an economy where fertility is exogenously fixed at $n = 1$.

It may seem counterintuitive that reducing financial barriers relative to the baseline leads to a slight decline in entrepreneurship, together with higher inequality and lower mobility.

The key mechanism is general equilibrium adjustment. Starting from an already relatively well-functioning financial market, a further reduction in financial barriers allows highly talented individuals to operate at a larger scale. Their increased demand for capital pushes up the equilibrium rental rate of capital. At the same time, as more resources are reallocated toward high-productivity entrepreneurs, less productive firms become less profitable and some exit entrepreneurship. The higher return to wealth also raises the income of asset-rich households. As a result, aggregate entrepreneurship falls slightly, while inequality rises and mobility declines.

Figures 14 and 15 illustrate the equilibrium price adjustments behind this mechanism. Figure 14 shows that reducing financial barriers shifts capital excess demand upward. The new equilibrium, therefore, requires a higher rental rate of capital. Figure 15 shows the corresponding adjustment in the labor market and labor excess demand shifts downward, implying a lower equilibrium wage. The non-smooth shape of the labor excess demand reflects the discrete nature of occupational choice. As wages change, some individuals cross occupational-choice thresholds and switch discretely between entrepreneurship and work. These threshold effects generate jumps or kinks in aggregate labor excess demand as a function of the wage.

The occupational responses are shown more directly in Figure 16. When financial barriers are reduced, constrained entrepreneurs can operate at their optimal scale. However, the overall entrepreneurship rate declines because some low-productivity entrepreneurs are no longer profitable at the new equilibrium prices and exit entrepreneurship. The figure should be interpreted with the underlying distribution in mind. Since population density varies across the state space, the visual area of a region does not necessarily correspond to its population mass.

Finally, Figure 17 reports the percentage change in income when financial barriers are reduced. The gains are concentrated among high-talent individuals, who benefit from the

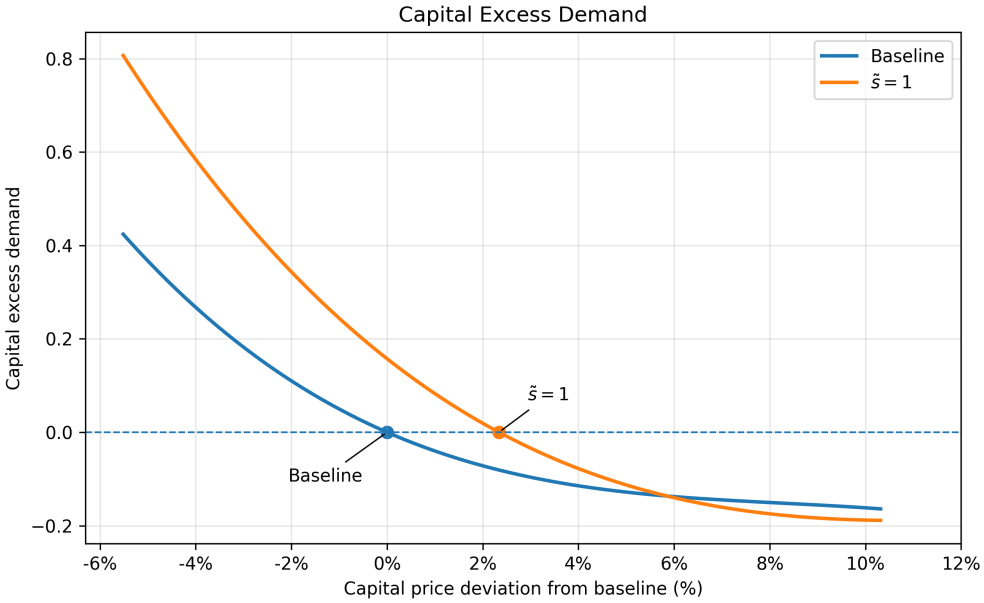


Figure 14: Capital Excess Demand: Baseline and When $\tilde{s} = 1$

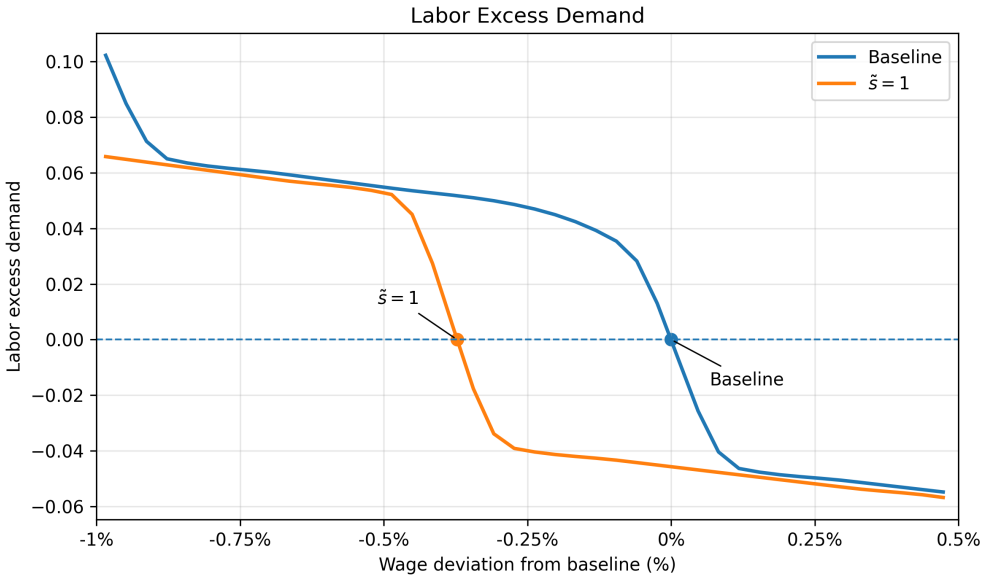
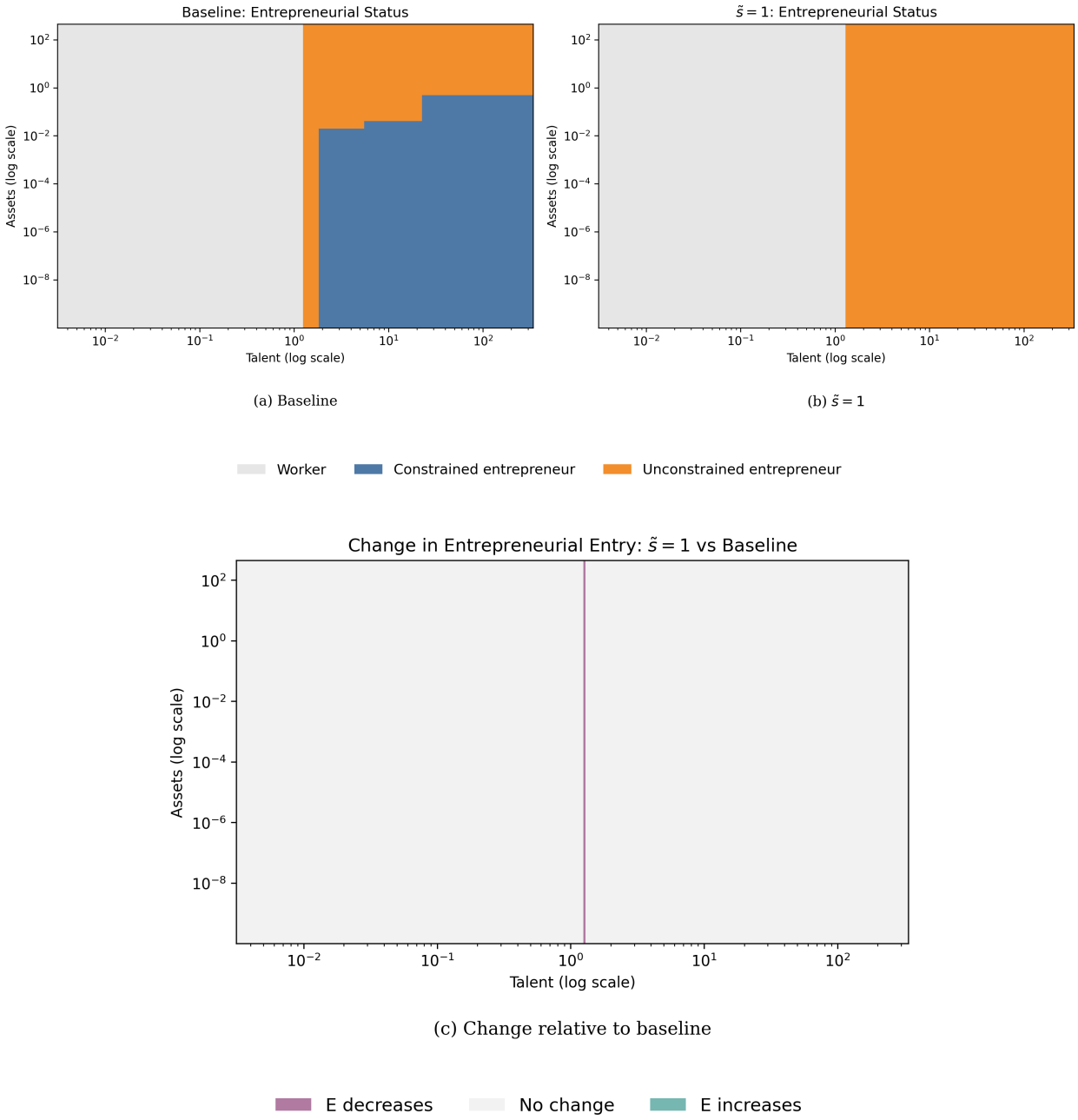


Figure 15: Labor Excess Demand: Baseline and When $\tilde{s} = 1$



(a) Baseline

(b) $\tilde{s} = 1$

(c) Change relative to baseline

Figure 16: Entrepreneurial status under the baseline and $\tilde{s} = 1$ cases

Notes: Panel (a) shows entrepreneurial status in the baseline economy. Panel (b) shows entrepreneurial status when $\tilde{s} = 1$. Panel (c) shows the change in entrepreneurial entry under $\tilde{s} = 1$ relative to the baseline. Entrepreneurial status is computed using Gauss–Legendre quadrature with 9 nodes within each cell. Talent and assets are plotted on log scales. Cell densities are not uniform across the state space, and the area of a region in the figure should not be interpreted as its population mass.

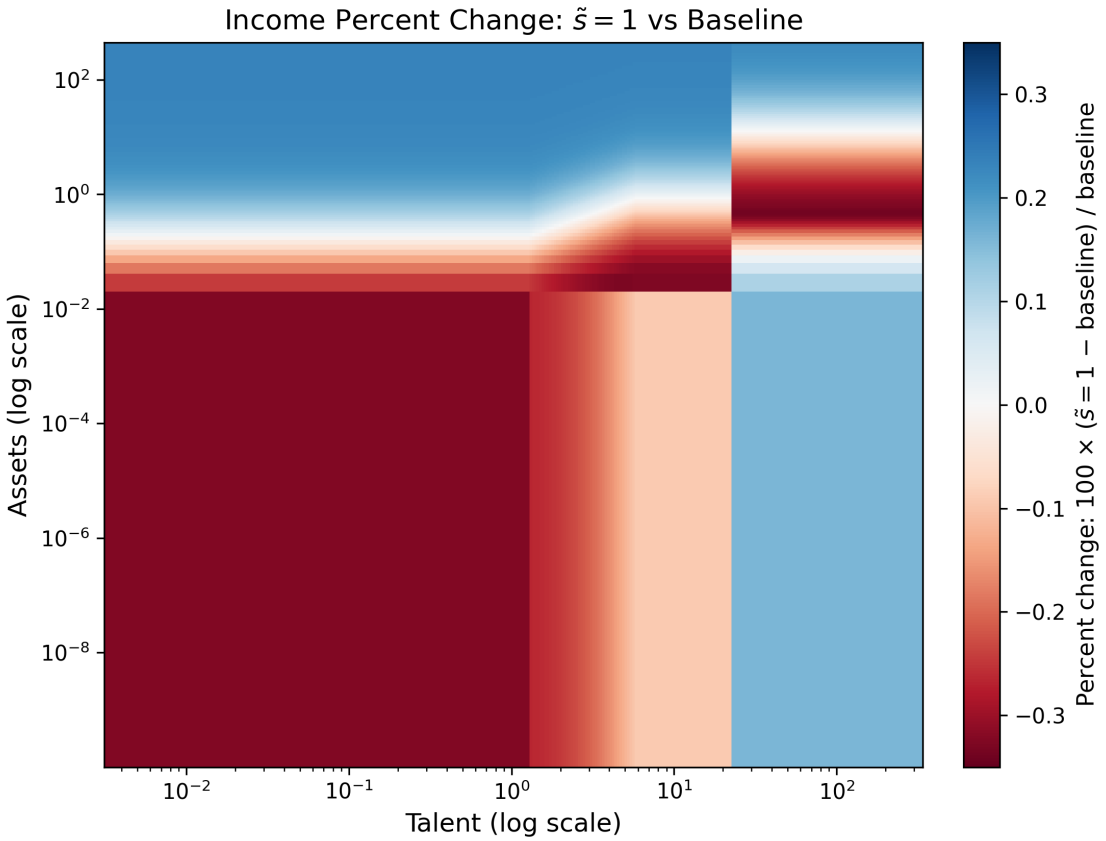


Figure 17: Income change when $\tilde{s} = 1$ relative to baseline

ability to run larger firms, and among high-asset individuals, who benefit from the higher return to capital. By contrast, individuals who rely more heavily on labor income are adversely affected by the decline in wages. This distributional pattern explains why a reduction in financial barriers can increase aggregate efficiency while simultaneously raising inequality and reducing mobility.

6.2 Shutting Down the Fertility Channel

I next examine the role of the endogenous fertility channel by comparing the endogenous-fertility economies in the first three columns of Table 4 with the fixed-fertility economies in the last three columns. In the fixed-fertility counterfactual, fertility is exogenously set to $n = 1$ under each level of financial barriers. This exercise shows how the quantitative

implications of financial barriers would change if fertility responses were ignored.

The comparison shows that shutting down the fertility channel changes both aggregate outcomes and distributional outcomes. Fixed fertility generally leads to higher entrepreneurship rates than endogenous fertility. The difference is small in the baseline economy and under $\tilde{s} = 1$, but it becomes much larger when financial barriers are tightened to $\tilde{s} = 0$: the entrepreneurship rate rises from 0.114 under endogenous fertility to 0.170 under fixed fertility. Thus, ignoring fertility responses would overstate entrepreneurship, especially when financial barriers are severe.

Fixed fertility also changes the estimated inequality and mobility. In the baseline economy and under $\tilde{s} = 1$, fixing fertility at $n = 1$ overstates both inequality and intergenerational mobility. For example, the Gini coefficient rises from 0.560 to 0.623 in the baseline economy and from 0.609 to 0.627 under $\tilde{s} = 1$. Intergenerational mobility also increases from 0.119 to 0.167 in the baseline economy and from 0.115 to 0.152 under $\tilde{s} = 1$. However, when financial barriers are tightened to $\tilde{s} = 0$, fixed fertility slightly understates the Gini coefficient and generates an even lower level of mobility. These results suggest that fertility choices interact with financial barriers in a nonlinear way.

The effect on aggregate development and welfare is also substantial. In the baseline economy and under $\tilde{s} = 1$, fixed fertility leads to much higher income per capita, consumption, and utility than in the endogenous-fertility economy. However, under $\tilde{s} = 0$, fixed fertility lowers income per capita relative to the endogenous-fertility economy, even though it raises consumption and utility. This pattern shows that fertility choices do not simply shift all outcomes in the same direction. Instead, they change how the economy adjusts to financial barriers.

These results have important implications for policy evaluation. A model that abstracts from endogenous fertility may mismeasure the effects of financial-market policies on en-

trepreneurship, inequality, mobility, and welfare. In particular, policies that change access to capital can affect the economy's demographics, which in turn feed back into aggregate and distributional economic outcomes. Ignoring this channel may therefore bias the evaluation of financial policies.

7 Conclusion

This paper examines how income-driven parental choices regarding fertility and bequests impact children's entry into entrepreneurship when financial barriers exist. To study these dynamics, I propose an OLG model with endogenous fertility, bequest, and occupational choices, and calibrate the model using US data. The model indicates that the interaction between parental decisions and borrowing constraints could lead to the misallocation of entrepreneurial talent and capital. Counterfactual analysis indicates that reducing financial barriers relative to the baseline model would increase income per capita but slightly reduce entrepreneurship due to the increased interest rate, thereby increasing inequality and reducing mobility. These findings underscore the importance of designing financial policies that promote efficiency while also addressing concerns about income concentration.

While the model allows for intergenerational persistence in entrepreneurial talent through an AR(1) process, it abstracts from endogenous parental human capital investment decisions that may also shape children's entrepreneurship potentials. Extending the framework to incorporate such investment choices would be a useful direction for future research and could further clarify how financial and human capital channels jointly affect entrepreneurship and intergenerational mobility.

In addition, although the quantitative analysis is calibrated to the U.S. economy, the mechanism studied here may be even more salient in countries at earlier stages of financial

development, where entrepreneurial entry depends more heavily on family resources. Extending the analysis to such settings would be another useful direction for future research and could shed further light on how family resources shape entrepreneurship and intergenerational mobility differently across different stages of economic and financial development.

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Appendix

A Theoretical Proof

A.1 Benchmark Model without Financial Barriers

A.1.1 The Entrepreneur's Problem

Assume the depreciation rate is 1. After each time period, all of the capital borrowed will be utilized and will not be transferred to the next period. As an entrepreneur, the individual needs to solve the following problem in the per capita setup:

$$\max_{k_{it},l_{it}} \pi_{it} = (Az_{it})^{1-\alpha-\beta}(k_{it})^{\alpha} l_{it}^{\beta} - R_t k_{it} - W_t l_{it}$$

z_{it} is the entrepreneurial talent of the individual. The first-order conditions of the problem are given by:

$$(Az_{it})^{1-\alpha-\beta}\alpha(k_{it})^{\alpha-1} l_{it}^{\beta} = R_t \tag{35}$$

$$(Az_{it})^{1-\alpha-\beta}(k_{it})^{\alpha}\beta l_{it}^{\beta-1} = W_t \tag{36}$$

$$\frac{k_{it}}{l_{it}} = \frac{\alpha W_t}{\beta R_t} \tag{37}$$

The optimal choices are:

$$k_{it}^{*} = (\frac{\alpha}{R_t})^{\frac{1-\beta}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} A z_{it} \tag{38}$$

$$l_{it}^{*} = (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{1-\alpha}{1-\alpha-\beta}} A z_{it} \tag{39}$$

The maximized firm profit is:

$$A z_{it} (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} (1-\alpha-\beta)$$

A.1.2 Entrepreneurship Decision

After evaluating the profit of starting a business, the individual will then try to compare the income of an entrepreneur and a worker to choose the occupation that will bring the highest income. The household chooses to start a firm if and only if the income from being an entrepreneur exceeds the income from working:

$$A z_{it} (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} (1-\alpha-\beta) + R_t a_{it} \geq W_t + R_t a_{it}$$

After solving the problem, we get the entrepreneurial cutoff. The individual only chooses to become an entrepreneur if their talent is higher than the threshold.

$$A z_{it} \geq \frac{\beta}{1-\alpha-\beta} (\frac{R_t}{\alpha})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{W_t}{\beta})^{\frac{1-\alpha}{1-\alpha-\beta}} \tag{40}$$

Define the entrepreneurial threshold as z_t^e. Note that in the benchmark model, the cutoff is universal across individuals.

Let's also define the Probability Density Function and Cumulative Distribution Function of entrepreneurial talent as g(z) and G(z), respectively. Moreover, I also define the average talent of entrepreneurs as $\phi(z)$:

$$\phi(z) = \int_{z_{it} \geq z_t^e} z_{it} g(z_{it}) dz_{it}$$

Now we have:

$$\frac{1}{N_t} \int_{i_t \in E} z_{it} d_{i_t} = E(z_{it} | z_{it} \geq z_t^e) \times (1 - G(z_t^e)) = \int_{z_{it} \geq z_t^e} z_{it} g(z_{it}) dz_{it} = \phi_t \tag{41}$$

A.1.3 Labor Market

From equation (39) and (41), we can calculate the labor demand per capita as follows:

$$l_t^d = (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{1-\alpha}{1-\alpha-\beta}} A \phi_t \tag{42}$$

As for labor supply per capita, I assume every adult worker provides one unit of labor inelastically. The market-clearing condition would be:

$$1 = (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{1-\alpha}{1-\alpha-\beta}} A \phi_t \tag{43}$$

A.1.4 Capital Market

According to equation (38), capital demand per capita is given by:

$$\int_{i_t \in \{E\}} k_{it}^s df(a_{it}) = (\frac{\alpha}{R_t})^{\frac{1-\beta}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} A\phi_t \tag{44}$$

The next step is to calculate the capital supply:

$$k_t^s = \int_{i_t} a_{it} df(a_{it}) \tag{45}$$

The clearing condition for the capital market is:

$$\int_{i_t} a_{it} f(a_{it}) = (\frac{\alpha}{R_t})^{\frac{1-\beta}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} A\phi_t \tag{46}$$

A.1.5 Equilibrium

In the benchmark model, we have the market-clearing condition for labor and capital from equation (43) and equation (46) as well as the relationship between R_t and W_t from equation (37). So now we have:

$$\frac{\alpha W_t}{\beta R_t} = \frac{\int_{i_t} a_{it} df(a_{it})}{G(z_t^e) N_t} = \frac{\overline{k}_t}{G(z_t^e)} \tag{47}$$

$$R_t = \alpha (\frac{1}{k_t})^{1-\alpha} (G_t)^{\beta} (A\phi_t)^{1-\alpha-\beta} \tag{48}$$

$$W_t = \beta(k_t)^{\alpha}(\frac{1}{G_t})^{1-\beta}(A\phi_t)^{1-\alpha-\beta} \tag{49}$$

k_t is the equilibrium capital per capita level.

A.1.6 Solution Existence & Uniqueness of z_t^e

Now using equation (48) and equation (49) we can rewrite the entrepreneurial cutoff as follows:

$$z_t^e = \frac{\beta}{1-\alpha-\beta}\frac{\phi_t}{G_t} \tag{50}$$

Define:

$$f(z_t^e) = \beta\frac{\phi_t}{G_t}$$

$$j(z_t^e) = (1-\alpha-\beta)z_t^e$$

We need to show that there is one unique z_t^e which could solve the following equation:

$$j(z_t^e) = f(z_t^e) \tag{51}$$

It's easy to show that:

$$\phi_t'(z_t^e) < 0$$

$$G_t'(z_t^e) > 0$$

As a result $f(z_t^e)$ is decreasing in z_t^e and $j(z_t^e)$ is increasing in z_t^e. When $z_t^e \to 0$, $G_t \to 0$, ϕ_t converges to a constant, $f(z_t^e)$ converges to infinity. When $z_t^e \to \infty$, $G_t \to 1$, $\phi_t \to 0$, $f(z_t^e) \to 0$. According to the Intermediate Value Theorem, there must be one unique z_t^e which could make equation (51) hold.

A.1.7 Law of Motion for Capital

Next, consider the law of motion for capital. According to the solution of the utility maximization problem, we know the optimal solution to total bequest m_{it} and fertility n_{it} are given by:

$$m_{it}^* = \frac{(\frac{\theta_2}{\theta_1})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it}$$

$$n_{it}^* = \frac{(\frac{1-\theta_1-\theta_2}{\theta_1 \tau})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it}^{1-\epsilon}$$

Accordingly, the capital per capita next generation should be defined as:

$$k_{t+1} = \frac{\int_{i_t} m_{it} di_t}{\int_{i_t} n_{it} di_t} = \frac{\int_{i_t} \frac{(\frac{\theta_2}{\theta_1})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it} \, di_t}{\int_{i_t} \frac{(\frac{1-\theta_1-\theta_2}{\theta_1 \tau})^\epsilon}{1 + (\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it}^{1-\epsilon} \, di_t} \tag{52}$$

The capital per capita of the next generation is purely decided by the income distribution of the current generation.

A.2 The Model with Financial Barriers

A.2.1 The Bank's Problem

First, we need to introduce the bank's problem when individuals may default after borrowing money from the bank. Banks are in a competitive market. They aim to lend as much capital as possible without making the entrepreneurs default.

$$\max_{k_{it}} \quad R_t(\int_{i_t \in E} k_{it} d_{it} - \int_{i_t} a_{it} di_t)$$

However, given the imperfect contract enforceability determined by the parameter s, entrepreneurs can renege on their contracts and keep a fraction $(1-s)$ of the profit without paying the capital interest. The bank sets the capital limit to prevent entrepreneurs from defaulting. Entrepreneurs do not default if and only if:

$$\begin{aligned} &\max_{k_{it} \in [0, k_{it}^{limit}]} (1-s)\{(Az_{it})^{1-\alpha-\beta}(k_{it})^{\alpha} l_{it}^{\beta} - W_t l_{it}\} - a_{it} \\ &\leq \max_{k_{it} \in [0, k_{it}^{limit}]} \{(Az_{it})^{1-\alpha-\beta}(k_{it})^{\alpha} l_{it}^{\beta} - R_t k_{it} - W_t l_{it}\} + R_t a_{it} \end{aligned} \tag{53}$$

We know that from firm's problem, given any k_{it}, the optimal level of labor should be:

$$\bar{l}_{it} = (\frac{\beta z_{it}^{1-\alpha-\beta}(k_{it})^{\alpha}}{W_t})^{\frac{1}{1-\beta}} \tag{54}$$

So we can rewrite problem (53) as follows:

$$\begin{aligned} &\max_{k_{it} \in [0, k_{it}^{limit}]} (1-s)(1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} - a_{it} \\ &\leq \max_{k_{it} \in [0, k_{it}^{limit}]} (1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} + R_t(a_{it} - k_{it}) \end{aligned} \tag{55}$$

Now the bank's problem becomes:

$$\max_{k_{it}^{limit}} \quad R_t(\int_{i_t \in E} k_{it}^{limit} d_{it} - \int_{i_t} a_{it} di_t)$$

Subject to:

$$\max_{k_{it} \in [0, k_{it}^{limit}]} (1-s)(1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} - a_{it}$$
$$\leq \max_{k_{it} \in [0, k_{it}^{limit}]} (1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} + R_t(a_{it} - k_{it})$$

A.2.2 Entrepreneurs' Borrowing Constraints

After solving the bank's problem, there will be two types of individuals. The first group of people can borrow enough money from the bank to operate the firm at their optimal capital level, so they are unbounded. The other group of individuals' optimal capital is lower than the capital they could borrow, so they are bound by the bank and can only produce at a suboptimal level.

For unbounded individuals, they produce at their optimal capital level as they do in the benchmark model. Their entrepreneurship cutoff is given by equation (40). If the bank wants to prevent them from defaulting

$$\max_{k_{it} \in [0, k_{it}^{limit}]} (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (1-s)(1-\beta) - a_{it} \leq$$
$$(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{*\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (1-\beta) + R_t(a_{it} - k_{it}^*)$$

$$\Longleftrightarrow$$

$$\max_{k_{it}\in[0,k_{it}^{limit}]}(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}}(k_{it}^{limit})^{\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}(1-s)(1-\beta)\leq$$
$$(1-\alpha-\beta)(Az_{it})(\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}}+(1+R_t)a_{it}$$

The bank wants to lend as much capital as possible, so the k_{it}^{limit} should be the solution to the following equation:

$$\begin{aligned}&(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}}(k_{it}^{limit})^{\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}(1-s)(1-\beta)=\\&(1-\alpha-\beta)(Az_{it})(\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}}+(1+R_t)a_{it}\end{aligned}\tag{56}$$

After solving (56), we have:

$$\begin{aligned}k_{it}^{limit}&=\left(\frac{(1-\alpha-\beta)(Az_{it})(\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}}+(1+R_t)b_{it-1}}{(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}(1-s)(1-\beta)}\right)^{\frac{1-\beta}{\alpha}}\\&=\left(\frac{\frac{1-\alpha-\beta}{\alpha}R_tk_{it}^*+(1+R_t)b_{it-1}}{\frac{(1-s)(1-\beta)}{\alpha}R_t(k_{it}^*)^{\frac{1-\alpha-\beta}{1-\beta}}}\right)^{\frac{1-\beta}{\alpha}}\end{aligned}$$

Since the individuals are unbounded, we require $k_{it}^*\leq k_{it}^{limit}$. To make $k_{it}^*\leq k_{it}^{limit}$:

$$\frac{s(1-\beta)-\alpha}{\alpha}R_tk_{it}^*+(1+R_t)a_{it}\geq 0\tag{57}$$

If $s\geq\frac{\alpha}{1-\beta}$, which means financial barriers are small, everyone who passes the entrepreneurial cutoff can get enough capital, and we are back to the benchmark model. So to make sure there exists a separating equilibrium, we also require:

$$s<\frac{\alpha}{1-\beta}\tag{58}$$

What's more, only individuals who satisfy the following condition will be unbounded.

$$\frac{a_{it}}{k_{it}^*} \geq \frac{\alpha - s(1-\beta)}{\alpha} \frac{R_t}{1+R_t} \tag{59}$$

Or equivalently, we can rewrite (59) as

$$\frac{a_{it}}{Az_{it}} \geq \frac{\alpha - s(1-\beta)}{\alpha} \frac{R_t}{1+R_t} (\frac{\alpha}{R_t})^{\frac{1-\beta}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} \tag{60}$$

The intuition behind this is that when the individual has more assets than the entrepreneurial talent, the individual will be unbounded. Next, let's discuss the case when the individual is bound by the bank. In the bound case, we can rewrite the equation (53) as follows:

$$\begin{aligned} &(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (1-s)(1-\beta) - a_{it} \leq \\ &(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (1-\beta) + R_t(a_{it} - k_{it}^{limit}) \end{aligned} \tag{61}$$

The capital limit k_{it}^{limit} is given by the following equation implicitly:

$$s(1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} - Rk_{it}^{limit} + (1+R_t)a_{it} = 0 \tag{62}$$

It's easy to show that k_{it}^{limit} is increasing and concave in both a_{it} and z_{it}.

For the bounded individuals, their entrepreneurial cutoff can be calculated by the following inequality given k_{it}^{limit}:

$$(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}} k_{it}^{limit\,\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (1-\beta) - R_t k_{it}^{limit} - W_t \geq 0 \tag{63}$$

After solving (63), we know that the bounded individuals will choose to become entrepreneurs if and only if:

$$Az_{it} \geq \left(\frac{W_t + R_t k_{it}^{limit}}{(1-\beta)(k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}} \right)^{\frac{1-\beta}{1-\alpha-\beta}} \tag{64}$$

A.2.3 Solution Existence & Uniqueness of Entrepreneurial Cutoff for the Bound

If we take the derivative of the right-hand side of (64), we should have:

$$\frac{(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} (k_{it}^{limit})^{\frac{\alpha}{1-\beta}-1}}{((1-\beta)(k_{it}^{limit})^{\frac{\alpha}{1-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}})^2} \times ((1-\alpha-\beta)\, R_t k_{it}^{limit} - \alpha W_t)$$

When $k_{it}^{limit} \geq \frac{\alpha}{1-\alpha-\beta} \frac{W_t}{R_t}$, the right-hand side increases in k_{it}^{limit}. When $k_{it}^{limit} < \frac{\alpha}{1-\alpha-\beta} \frac{W_t}{R_t}$, the right-hand side decreases in k_{it}^{limit}. In the limit, as $z_{it} \to 0$, k_{it}^{limit} converges to a strictly positive constant, and the right-hand side converges to a strictly positive constant. As $z_{it} \to \infty$, k_{it}^{limit} again converges to a strictly positive constant, while the right-hand side remains finite. Therefore, the equation has a unique solution.

The law of motion for capital per capita remains the same for the case when there is no financial barrier.